

What is a Transistor?

A transistor is a semiconductor device that transfers a weak signal from a low resistance circuit to a high resistance circuit. In simple words, what it means is that it regulates and amplifies electrical signals such as voltage or current.

Transistors are special because they allow you to control how much current flows through a circuit. This can be achieved by controlling the voltage across two of the transistor leads. Each transistor has three leads.

Classification of Transistors According to Construction

Transistors are categorized into two types based on their construction, Bipolar Junction Transistors (BJT) and Field Effect Transistors (FET).

What are Bipolar Junction Transistors?

Bipolar Junction Transistors are also known as junction transistors. These were the first type of transistors that were mass-produced in 1947 by Bell Labs. These transistors are a combination of two junction diodes.

The three-layer structure of the junction transistors can contain either of the combinations:

- An n-type semiconductor layer sandwiched between p-type layers forming a p-n-p configuration

or

- A p-type layer between n-type layers forming an n-p-n configuration.

It has two junctions between p-type and n-type semiconductors. BJTs are current-controlled devices which mean that a small amount of current flowing through the base of a Bipolar Junction Transistor results in a large current that flows from emitter to collector.

What is Field Effect Transistors (FET)?

A field-effect transistor uses an electric field to control the flow of current. They have three terminals which are named *source*, *gate*, and *drain*. FETs control the flow of current by the application of a voltage to the gate, which in turn alters the conductivity between the drain and source. FETs are classified into three types as Junction type FETs, MOS (Metal-Oxide-Semiconductor) type FETs, and MES (Metal-Semiconductor) type FETs. Junction type FETs are used in analog circuits such as those in audio equipment. MOS type FETs are used in digital ICs such as those used in microcomputers. MES type FETs are used for the amplification of microwaves.

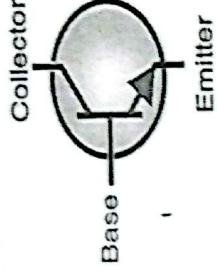
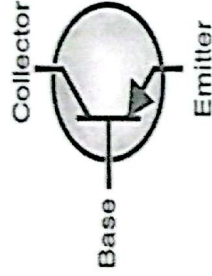
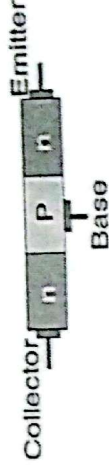
Construction of Bipolar Junction Transistor

A transistor is a three-layer semiconductor device in which one type of semiconductor (either P-type or N-type) is sandwiched between two other similar types of semiconductors. A bipolar junction transistor is formed by three layers of semiconductor materials, if it is a p-n-p transistor, it will have two p-type regions and one n-type region, likewise, if it is an n-p-n transistor, it will have two n-type regions and one p-type region.

PNP



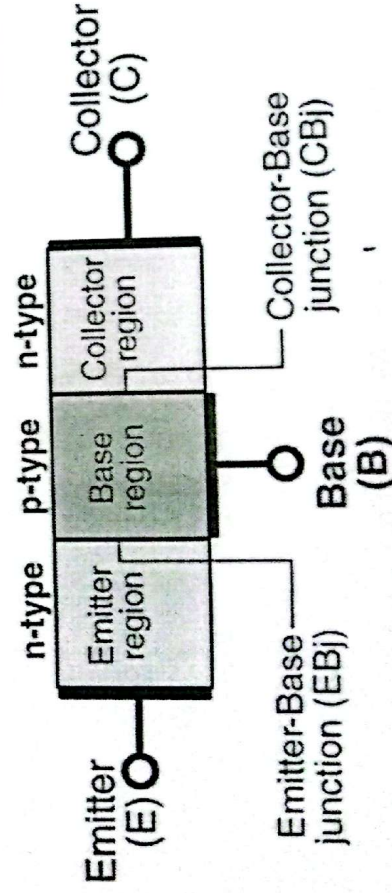
NPN



Transistors have three terminals namely emitter, collector and base. We have explained the functionalities of each of these terminals below:

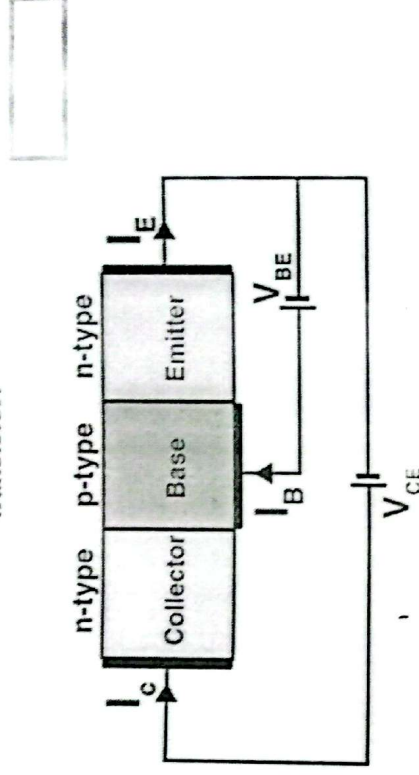
- **Emitter** – In a transistor, the emitter supplies a large section of majority charge carriers. The emitter is always forward biased with respect to the base so that it supplies the majority charge carrier to the base. The emitter of a transistor is heavily doped and moderate in size.
- **Collector** – In a transistor, the section that collects the majority of the charge carrier supplied by the emitter is called a collector. The collector-base junction is always reverse biased. The collector section of the transistor is moderately doped, but larger in size so that it can collect most of the charge carrier supplied by the emitter.
- **Base** – The middle section of the transistor is known as the base. The base forms two circuits, the input circuit with the emitter and the output circuit with the collector. The emitter-base junction is forward-biased and offers low resistance to the circuit. The collector-base junction is in reverse bias and offers higher resistance to the circuit. The base of a transistor is lightly doped and very thin due to which it offers the majority charge carrier to the base.

The Action of n-p-n Transistor



The n-p-n transistor consists of two n-type semiconductors that sandwich a p-type semiconductor. Here, electrons are the majority charge carriers while holes are the minority charge carriers.

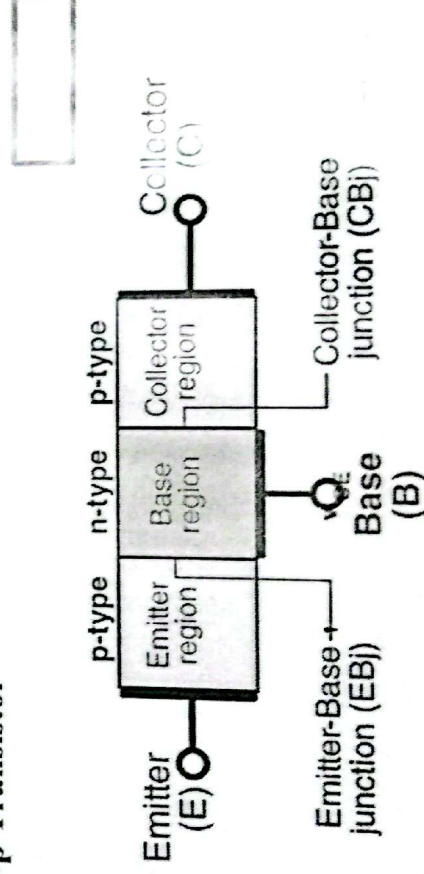
In an n-p-n transistor, the majority of the charge carriers are electrons and holes are the minority charge carriers. A small amount of current at the base terminal causes a large amount of current to flow from emitter to collector. The figure below represents the circuit diagram of the n-p-n transistor:



From the circuit diagram of the n-p-n transistor, it is seen that the emitter-base circuit is forward biased while the collector-emitter circuit is reverse biased.

Due to the forward bias, the majority of charge carriers in the emitter are repelled towards the base. The electron-hole recombination is very small in the base region because the base is lightly doped. Most of the electrons cross into the collector region.

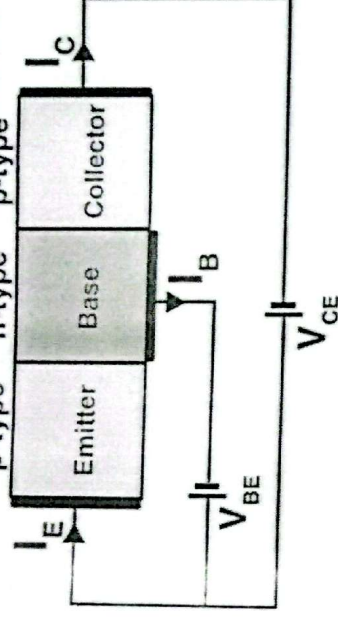
The Action of p-n-p Transistor



The p-n-p transistor consists of two p-type semiconductors that sandwich an n-type semiconductor. Here, holes are the majority charge carriers while electrons are the minority charge carriers.

The figure below represents the circuit diagram of the p-n-p transistor:

2.3.1.1. **Common Emitter (CE) Configuration** – In this configuration, the input signal is applied to the base terminal, the output is taken from the collector terminal, and the emitter terminal is connected to ground. This configuration provides a voltage gain and a phase shift of 180 degrees. The collector current is approximately equal to the emitter current, and the base current is much smaller. The input impedance is moderate, and the output impedance is high. The common emitter configuration is the most widely used transistor configuration.



The emitter-base (V_{BE}) battery connects the p-type emitter which is forward biased whereas the collector-base (V_{BC}) battery connects the p-type collector which is reverse biased. In this case, the majority charge carriers in emitter are holes which are repelled towards the base. As the base layer is thin, thus only little interaction occurs when electrons and holes combine. Most of the holes reach the collector. The current is carried by holes in p-n-p transistors.

Why are n-p-n Transistors Most Preferred Over p-n-p Transistors
There are several reasons for this:

- **Carrier Mobility** – The majority charge carriers in n-p-n transistors are electrons unlike in p-n-p transistors where the majority charge carriers are holes. Electrons move far more easily than holes within the crystal lattice. As a result, they have higher mobility and operate faster providing a much better level of performance.
- **Production Costs** – The manufacture of silicon-based transistors is most economically carried out using large N-type silicon wafers. The manufacture of PNP transistors requires three times more surface area of the wafer, and this significantly increases the costs.
- **Negative Grounding** – Over the years, a negative ground has become standard and the polarity of NPN transistors means that the basic transistor configurations operate with a negative ground.

Transistor Modes of Operation

A transistor consists of two junctions which can be biased in different ways. The different working modes of the transistor based on different junction biasing are given in the table below:

Condition	Emitter Junction (EB)	Collector Junction (CB)	Region of Operation
FR	Forward-biased	Reversed-biased	Active
FF	Forward-biased	Forward-biased	Saturation
RR	Reversed-biased	Reversed-biased	Cut-off
RF	Reversed-biased	Forward-biased	Inverted

FR – In this case, the emitter-base junction is forward biased and the collector-base junction is reverse biased. The transistor is in the active region and the collector current depends on the emitter current. The transistor which operates in this region is used for amplification.

FF – In this case, both the junctions are forward biased. The transistor is in saturation and the collector current becomes independent of the base current. The transistors act like a closed switch.

RR – In this case, both the junctions are reverse biased. The emitter does not supply the majority charge carriers to the base because of which the carrier current is not collected by the collector. Thus, transistors, in this case, act like an open switch.

RF – In this case, the emitter-base junction is reverse biased and the collector-base junction is forward biased. As the collector is lightly doped compared to the emitter junction it does not supply the majority charge carrier to the base. Hence proper transistor action is achieved.

Difference Between NPN and PNP Transistor	
NPN	PNP
The current flows from collector terminal to emitter terminal.	The current flows from emitter to collector terminal.
One P-type semiconductor is sandwiched between the two N-type semiconductors.	It is made of up two P-type material layers with N-type sandwiched between them.
The current flow from the collector is generated by keeping a +ve voltage there.	The current flow from the emitter to collector is generated at emitter terminal by keeping a +ve voltage there.
The transistor switches ON with the increase in current in the base terminal	The transistors switch ON when there is no current flow at the base terminal
When the current is reduced in the base, the transistor doesn't function across the collector terminal and switches OFF	When a current is present at the base of a PNP transistor, then the transistor switches OFF.

FR – In this case, the emitter-base junction is forward biased and the collector-base junction is reverse biased. The transistor is in the active region and the collector current depends on the emitter current. The transistor which operates in this region is used for amplification.

FF – In this case, both the junctions are forward biased. The transistor is in saturation and the collector current becomes independent of the base current. The transistors act like a closed switch.

RR – In this case, both the junctions are reverse biased. The emitter does not supply the majority charge carriers to the base because of which the carrier current is not collected by the collector. Thus, transistors, in this case, act like an open switch.

RF – In this case, the emitter-base junction is reverse biased and the collector-base junction forward biased. As the collector is lightly doped compared to the emitter junction it does not supply the majority charge carrier to the base. Hence poor transistor action is achieved.